

Sarang N S

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PROFESSIONAL PROFILE

Pre-final year ECE student at SVNIT (9.43 CGPA, Minor in AI) with hands-on experience in ML pipeline engineering, model optimization, and end-to-end system development. Research Intern at IIT Gandhinagar who built ML surrogate models with synthetic data generation, neural network training, ONNX export, and Unity Sentis integration for real-time inference. Co-authored a paper accepted at **CVPR 2026** (NTIRE Workshop). Passionate about Edge AI and the intersection of machine learning with hardware systems – driven to bridge the gap between powerful ML models and the resource-constrained devices they need to run on efficiently in the real world.

EDUCATION

B.Tech, ECE (Minor: AI)	SVNIT, Surat	2024 – Present	9.43 CGPA
Class 12, CBSE	Sri Chaithanya Techno School	2024	95.6%
Class 10, CBSE	Royal Concorde International School	2022	97.0%

RESEARCH EXPERIENCE

Research Intern, SRIP 2026 – ML for Semiconductor Process Modeling	IIT Gandhinagar
<i>V-FAB 1.0: ML-Driven Interactive Semiconductor Education App</i>	<i>May 2026 – July 2026</i>
- Surveyed primary literature to extract semi-empirical etch rate models across gas chemistries (CF₄, SF₆); fitted governing equations to generate a physically consistent synthetic training dataset. - Trained an XGBoost model with a FastAPI serving backend for initial inference; identified server startup latency as a bottleneck and migrated to a neural network exported via ONNX and Unity Sentis for direct on-engine inference – eliminating server dependency entirely. - Delivered a fully functional interactive semiconductor education app integrating real-time ML inference with 3D Unity/C# simulation, enabling students to visualize fabrication processes at an atomic level.	

PROJECTS

Energy-Aware CNN Optimization for Edge AI <i>PyTorch, Model Compression, Quantization</i>	GitHub
- Achieved 73% model compression and 4x energy reduction via Iterative Magnitude-based Pruning and INT8 Post-Training Static Quantization, maintaining viable predictive performance. - Built a Pareto-optimal benchmarking framework mapping Top-1 accuracy vs. model size vs. power envelope – demonstrating ML model optimization for resource-constrained AI hardware deployment.	
ASTRAM – Traffic Congestion Forecasting <i>LightGBM, XGBoost, Streamlit, Python</i>	GitHub
<i>Flipkart Gridlock Hackathon (Team Project)</i> - Engineered an end-to-end ML pipeline for traffic congestion severity forecasting across Bengaluru using LightGBM with SMOTE oversampling, achieving ~99.9% Weighted F1 score. - Tuned XGBoost classifier with decision threshold optimization (0.638) to maximize minority class recall on severely imbalanced road closure data (8.2% positive rate). - Deployed an interactive Streamlit dashboard integrating real-time predictions, hotspot maps, and resource recommendation outputs.	
Multi-Style Fast Neural Style Transfer <i>PyTorch, Perceptual Loss, Computer Vision</i>	GitHub
- Implemented real-time neural network stylization using a feed-forward perceptual loss network; achieved significant inference speedup over iterative optimization methods. - Applied Instance Normalization and Residual connections to stabilize training and preserve high-frequency structural detail during stylization.	

TECHNICAL SKILLS

Languages	Python, C/C++, C#
ML / DL Frameworks	PyTorch, Torchvision, OpenCV, Scikit-learn, LightGBM, XGBoost
ML Techniques	Deep Learning, Neural Network Training, Transfer Learning, CNNs, Model Compression, Quantization, Pruning, Data Preprocessing, Model Inference, Computer Vision
Data and Tools	NumPy, Pandas, Matplotlib, Git, FastAPI, Streamlit, Unity, Unity Sentis, ONNX
Relevant Coursework	Machine Learning, Computer Vision, Computer Architecture and Organization, OOPs in C++, Digital Logic Design

ACHIEVEMENTS AND COMPETITIONS

- Co-author, CVPR 2026 (NTIRE Workshop):** Co-authored MELD-SR, accepted at CVPR 2026 NTIRE Workshop – selected for methodological novelty in Image Super-Resolution x4, competing against teams globally.
- Flipkart Gridlock Hackathon 2.0:** Qualified for competitive national-level hackathon; developed ASTRAM, an end-to-end ML platform for traffic congestion forecasting.